CSCI-544 Research-Focused Project Proposal

Faithfulness MATTERS (Modality Attribution Testing Through Emotion Recognition Signals)

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What are you trying to do?
Multimodal emotion recognition (MER) systems combine what someone says with how they say it to predict emotion. Some MER systems even produce a score suggesting whether the words or tone mattered more in that specific prediction. We are trying to test whether that score is trustworthy: if a system says tone mattered more, does changing or removing the audio actually change its prediction more than changing the words? We plan to compare internal fusion-gate weights and per-modality confidence against measured changes in the model's predictions when text or audio is altered. We will build a small text+audio emotion classifier on MELD to test this.

How is it done today, and what are the limits of current practice?

- MELD is a standard text+audio+video benchmark for MER in conversation (*Poria et al. 2019*).
- MER systems can fuse modalities through methods such as learned gating (e.g. *He and Xiao 2025*), producing quantities such as gate weights that can be interpreted as modality importance.
- However, an importance score does not necessarily reflect actual model dependence; a high score for one modality does not by itself show that changing that modality would affect the prediction most.
- *Jacovi and Goldberg (2020)* distinguish plausible explanations from explanations that are faithful to model behavior, motivating our intervention-based evaluation.

What is new in your approach and why do you think it will be successful? We test faithfulness as consistency between what the model says is important and what actually affects its prediction, without claiming to identify the one true cause of a prediction. This should provide a useful test because the interventions directly measure how sensitive the model's prediction is to each modality rather than relying on the importance signal itself.

- **Core signals:** gate weight and text-only/audio-only confidence, optionally extended with gradient/SHAP-based importance scores.
- **Interventions:** the actual change in prediction when we remove or corrupt each modality.
- **Conflict test:** an utterance's text paired with audio from another utterance carrying a different MELD emotion label, to see whether consistency holds when modalities disagree, not only when they naturally agree.
- **Hypothesis:** agreement between what the model says is important and what actually affects its prediction will decrease when text and audio provide conflicting emotional evidence.
- **Metric:** modality-ranking agreement (whether both methods identify the same modality as more important) plus Spearman correlation between the two.
- **Scope:** We limit our conclusions to the model(s) we test, not to fusion gates in general.

Who cares? Anyone auditing an MER system, in assistants, social robots, content moderation, or upstream of generative dialogue systems (e.g. games or film), needs to know whether "the model relied on your tone" is true of its behavior or a plausible-sounding story: a passing test lets us treat the signal as a debugging tool, a failing test tells us not to.

What are the risks?

- MELD contains naturally paired text and audio rather than controlled text-audio conflict examples, so we build a synthetic conflict set by pairing an utterance's text with audio from another utterance carrying a different MELD emotion label.
- Removing a modality can create inputs unlike those the model normally sees. We compare two intervention methods, silence and mean-embedding replacement. If they disagree, we report that instability rather than treating either as reliable evidence of model dependence.

How much will it cost? Only lightweight classifier and gate components are trained on frozen, pretrained encoders, keeping the proposed experiments feasible within a semester on one GPU (USC's CARC Discovery cluster or equivalent), with no specialized hardware required for development.

Milestones:

- **Midterm:** unimodal and fused baselines reproduced on the official MELD splits, reporting macro- and weighted-F1, plus tests showing how removing text or audio affects the model's predictions.
- **Final:** full faithfulness comparison across importance signals, including the text-audio conflict examples.